

Supplementary File 2. Full Search Strategy

Title

Supplementary File 2. Database and Grey Literature Search Strategies

1. Purpose

This supplementary file documents the full search strategy for the scoping review titled **“Reframing the malaria-sickle cell disease relationship in sub-Saharan Africa as a bounded syndemic: A scoping review and evidence gap map.”** The purpose of this file is to make the search process transparent, reproducible, and auditable.

The search strategy was designed to identify evidence on malaria and sickle cell disease (SCD) relationships in sub-Saharan Africa (SSA), including biological interaction, red blood cell polymorphisms, spatial co-risk, clinical convergence, under-recognition, service-mediated visibility, structural vulnerability, newborn screening, screening-to-care linkage, and syndemic theory.

The search was structured around seven concept blocks:

1. Sickle cell disease, sickle cell anaemia, HbS, HbAS, and sickle cell trait
2. Malaria and *Plasmodium falciparum*
3. Red blood cell polymorphisms
4. Clinical expression, including severe anaemia and transfusion need
5. Recognition, attribution, routine data, and service visibility
6. Newborn screening, diagnosis, treatment, referral, and care linkage
7. Syndemic theory, structural vulnerability, health systems, and programme response

2. Search summary table

Source	Platform	Date searched	Search strategy version	Records retrieved	Notes
PubMed or MEDLINE	PubMed	To be inserted	Version 1.0	To be inserted	Includes MeSH and free-text terms
Embase	Elsevier or Ovid	To be inserted	Version 1.0	To be inserted	Includes Emtree and

					free-text terms
Scopus	Elsevier	To be inserted	Version 1.0	To be inserted	Title, abstract, keyword search
Web of Science Core Collection	Clarivate	To be inserted	Version 1.0	To be inserted	Topic search
Global Health	Ovid or EBSCO	To be inserted	Version 1.0	To be inserted	Subject headings and free-text terms
CINAHL	EBSCOhost	To be inserted	Version 1.0	To be inserted	CINAHL headings and free-text terms
African Index Medicus	WHO	To be inserted	Version 1.0	To be inserted	Simplified search syntax
Cochrane Library	Wiley	To be inserted	Version 1.0	To be inserted	Trials, reviews, protocols
Grey literature	Multiple sources	To be inserted	Version 1.0	To be inserted	WHO, Africa CDC, SickleInAfrica, ministries, programme reports
Citation searching	Google Scholar, database citation tools, reference lists	To be inserted	Version 1.0	To be inserted	Backward and forward citation searching

3. Core search concepts

3.1 SCD and HbS terms

“sickle cell disease” OR “sickle cell anaemia” OR “sickle cell anemia” OR SCD OR SCA OR HbSS OR HbAS OR “sickle cell trait” OR “haemoglobin S” OR “hemoglobin S” OR “Hb S” OR “sickle haemoglobin” OR “sickle hemoglobin” OR “haemoglobinopathy” OR “hemoglobinopathy”

3.2 Malaria terms

malaria OR “Plasmodium falciparum” OR falciparum OR parasitaemia OR parasitemia OR PfHRP2 OR “histidine-rich protein 2” OR “severe malaria” OR “cerebral malaria” OR “severe malarial anaemia” OR “severe malarial anemia” OR “malaria parasite density” OR “parasite biomass”

3.3 Red blood cell polymorphism terms

“red blood cell polymorphism” OR “erythrocyte polymorphism” OR “red cell polymorphism” OR “alpha thalassaemia” OR “alpha thalassemia” OR “alpha-plus-thalassaemia” OR “alpha-plus-thalassemia” OR G6PD OR “glucose-6-phosphate dehydrogenase” OR “haemoglobin C” OR “hemoglobin C” OR HbC OR ABO OR “blood group O” OR Dantu OR epistasis OR “erythrocyte membrane” OR “parasite invasion” OR rosetting OR cytoadherence

3.4 Clinical expression terms

“severe anaemia” OR “severe anemia” OR anaemia OR anemia OR transfusion OR “blood transfusion” OR “respiratory distress” OR “acute chest syndrome” OR fever OR infection OR “febrile illness” OR “malaria-like illness” OR “clinical phenotype” OR “severe illness” OR “paediatric admission” OR “pediatric admission” OR “hospital admission” OR mortality OR morbidity

3.5 Recognition and visibility terms

underdiagnosis OR under-recognition OR “under recognition” OR “missed diagnosis” OR “clinical recognition” OR misclassification OR attribution OR “diagnostic attribution” OR “routine data” OR “health information system” OR DHIS2 OR HMIS OR coding OR reporting OR “mortality attribution” OR “cause of death” OR “service-visible burden” OR surveillance OR registry OR register

3.6 Screening and care terms

“newborn screening” OR “neonatal screening” OR “infant screening” OR “point-of-care testing” OR “point of care testing” OR “confirmatory diagnosis” OR counselling OR counseling OR “caregiver notification” OR hydroxyurea OR hydroxycarbamide OR penicillin OR prophylaxis OR vaccination OR immunisation OR immunization OR transfusion OR referral OR retention OR “linkage to care” OR “continuum of care” OR “comprehensive care” OR “standardised care” OR “standardized care”

3.7 Syndemic and structural context terms

syndemic OR syndemics OR “structural vulnerability” OR “structural violence” OR “health system” OR “health systems” OR “service readiness” OR “spatial prioritisation” OR “spatial prioritization” OR “hub and spoke” OR “facility readiness” OR “access to care” OR “implementation science” OR “programme response” OR “program response” OR decentralisation OR decentralization OR “health service access”

3.8 Geography terms

“Africa South of the Sahara” OR “sub-Saharan Africa” OR SSA OR Africa OR Angola OR Benin OR Botswana OR “Burkina Faso” OR Burundi OR Cameroon OR “Cape Verde” OR “Cabo Verde” OR “Central African Republic” OR Chad OR Comoros OR Congo OR “Democratic Republic of Congo” OR DRC OR “Republic of Congo” OR “Côte d’Ivoire” OR “Cote d’Ivoire” OR Djibouti OR Eritrea OR Eswatini OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR “Guinea-Bissau” OR Kenya OR Lesotho OR Liberia OR Madagascar OR Malawi OR Mali OR Mauritania OR Mauritius OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR “Sao Tome and Principe” OR Senegal OR Seychelles OR “Sierra Leone” OR Somalia OR “South Africa” OR “South Sudan” OR Sudan OR Tanzania OR Togo OR Uganda OR Zambia OR Zimbabwe

4. PubMed or MEDLINE search strategy

Database: PubMed or MEDLINE

Date searched: To be inserted

Records retrieved: To be inserted

4.1 PubMed search string

```
(  
  "Anemia, Sickle Cell"[Mesh]  
  OR "Sickle Cell Trait"[Mesh]  
  OR "Hemoglobin, Sickle"[Mesh]  
  OR "sickle cell disease"[tiab]  
  OR "sickle cell anaemia"[tiab]  
  OR "sickle cell anemia"[tiab]  
  OR SCD[tiab]  
  OR SCA[tiab]  
  OR HbSS[tiab]  
  OR HbAS[tiab]  
  OR "sickle cell trait"[tiab]  
  OR "haemoglobin S"[tiab]  
  OR "hemoglobin S"[tiab]  
  OR "Hb S"[tiab]  
  OR "sickle haemoglobin"[tiab]  
  OR "sickle hemoglobin"[tiab]  
  OR haemoglobinopathy[tiab]  
  OR hemoglobinopathy[tiab]  
)
```

AND

(

"Malaria"[Mesh]
OR "Malaria, Falciparum"[Mesh]
OR malaria[tiab]
OR "Plasmodium falciparum"[tiab]
OR falciparum[tiab]
OR parasitaemia[tiab]
OR parasitemia[tiab]
OR PfHRP2[tiab]
OR "histidine-rich protein 2"[tiab]
OR "severe malaria"[tiab]
OR "cerebral malaria"[tiab]
OR "severe malarial anaemia"[tiab]
OR "severe malarial anemia"[tiab]
OR "malaria parasite density"[tiab]
OR "parasite biomass"[tiab]
OR "red blood cell polymorphism"[tiab]
OR "erythrocyte polymorphism"[tiab]
OR "red cell polymorphism"[tiab]
OR "alpha thalassaemia"[tiab]
OR "alpha thalassemia"[tiab]
OR "alpha-plus-thalassaemia"[tiab]
OR "alpha-plus-thalassemia"[tiab]
OR G6PD[tiab]
OR "glucose-6-phosphate dehydrogenase"[tiab]
OR "haemoglobin C"[tiab]
OR "hemoglobin C"[tiab]
OR HbC[tiab]
OR ABO[tiab]
OR "blood group O"[tiab]
OR Dantu[tiab]
OR epistasis[tiab]

)

AND

(

"Africa South of the Sahara"[Mesh]
OR "Africa"[Mesh]
OR "sub-Saharan Africa"[tiab]
OR "sub Saharan Africa"[tiab]
OR SSA[tiab]
OR Africa[tiab]
OR Angola[tiab]
OR Benin[tiab]
OR Botswana[tiab]
OR "Burkina Faso"[tiab]

OR Burundi[tiab]
OR Cameroon[tiab]
OR "Cape Verde"[tiab]
OR "Cabo Verde"[tiab]
OR "Central African Republic"[tiab]
OR Chad[tiab]
OR Comoros[tiab]
OR Congo[tiab]
OR "Democratic Republic of Congo"[tiab]
OR DRC[tiab]
OR "Republic of Congo"[tiab]
OR "Côte d'Ivoire"[tiab]
OR "Cote d'Ivoire"[tiab]
OR Djibouti[tiab]
OR Eritrea[tiab]
OR Eswatini[tiab]
OR Ethiopia[tiab]
OR Gabon[tiab]
OR Gambia[tiab]
OR Ghana[tiab]
OR Guinea[tiab]
OR "Guinea-Bissau"[tiab]
OR Kenya[tiab]
OR Lesotho[tiab]
OR Liberia[tiab]
OR Madagascar[tiab]
OR Malawi[tiab]
OR Mali[tiab]
OR Mauritania[tiab]
OR Mauritius[tiab]
OR Mozambique[tiab]
OR Namibia[tiab]
OR Niger[tiab]
OR Nigeria[tiab]
OR Rwanda[tiab]
OR "Sao Tome and Principe"[tiab]
OR Senegal[tiab]
OR Seychelles[tiab]
OR "Sierra Leone"[tiab]
OR Somalia[tiab]
OR "South Africa"[tiab]
OR "South Sudan"[tiab]
OR Sudan[tiab]
OR Tanzania[tiab]
OR Togo[tiab]
OR Uganda[tiab]

OR Zambia[tiab]
OR Zimbabwe[tiab]
)

4.2 PubMed Supplementary search for screening, care, visibility, and syndemic terms

This supplementary search is intended to capture SCD programme, screening, and visibility studies that may not mention malaria directly but are relevant to malaria-endemic SSA settings and the syndemic framework.

(
"Anemia, Sickle Cell"[Mesh]
OR "Sickle Cell Trait"[Mesh]
OR "sickle cell disease"[tiab]
OR "sickle cell anaemia"[tiab]
OR "sickle cell anemia"[tiab]
OR SCD[tiab]
OR SCA[tiab]
OR HbSS[tiab]
OR HbAS[tiab]
OR "sickle cell trait"[tiab]
OR "haemoglobin S"[tiab]
OR "hemoglobin S"[tiab]
)
AND
(
"newborn screening"[tiab]
OR "neonatal screening"[tiab]
OR "infant screening"[tiab]
OR "point-of-care testing"[tiab]
OR "point of care testing"[tiab]
OR "confirmatory diagnosis"[tiab]
OR counselling[tiab]
OR counseling[tiab]
OR "caregiver notification"[tiab]
OR hydroxyurea[tiab]
OR hydroxycarbamide[tiab]
OR penicillin[tiab]
OR prophylaxis[tiab]
OR vaccination[tiab]
OR immunisation[tiab]
OR immunization[tiab]
OR transfusion[tiab]
OR referral[tiab]
OR retention[tiab]
OR "linkage to care"[tiab]
OR "continuum of care"[tiab]
)

```

OR "comprehensive care"[tiab]
OR "routine data"[tiab]
OR "health information system"[tiab]
OR DHIS2[tiab]
OR HMIS[tiab]
OR underdiagnosis[tiab]
OR under-recognition[tiab]
OR "missed diagnosis"[tiab]
OR "mortality attribution"[tiab]
OR syndemic[tiab]
OR syndemics[tiab]
OR "structural vulnerability"[tiab]
OR "service readiness"[tiab]
)
AND
(
  "Africa South of the Sahara"[Mesh]
  OR "sub-Saharan Africa"[tiab]
  OR "sub Saharan Africa"[tiab]
  OR SSA[tiab]
  OR Africa[tiab]
  OR Angola[tiab]
  OR Benin[tiab]
  OR Botswana[tiab]
  OR "Burkina Faso"[tiab]
  OR Burundi[tiab]
  OR Cameroon[tiab]
  OR Ghana[tiab]
  OR Kenya[tiab]
  OR Malawi[tiab]
  OR Mali[tiab]
  OR Nigeria[tiab]
  OR Tanzania[tiab]
  OR Uganda[tiab]
  OR Zambia[tiab]
  OR Zimbabwe[tiab]
)

```

5. Embase search strategy

Database: Embase

Platform: Elsevier or Ovid, to be inserted

Date searched: To be inserted

Records retrieved: To be inserted

The Embase strategy should be adapted depending on the platform. The following syntax uses Ovid-style field tags and proximity operators.

1. exp sickle cell disease/ OR exp sickle cell trait/ OR exp hemoglobin S/
2. ("sickle cell disease" OR "sickle cell anaemia" OR "sickle cell anemia" OR SCD OR SCA OR HbSS OR HbAS OR "sickle cell trait" OR "haemoglobin S" OR "hemoglobin S" OR "Hb S" OR "sickle haemoglobin" OR "sickle hemoglobin" OR haemoglobinopathy OR hemoglobinopathy).ti,ab,kw.
3. 1 OR 2
4. exp malaria/ OR exp falciparum malaria/ OR exp Plasmodium falciparum/
5. (malaria OR "Plasmodium falciparum" OR falciparum OR parasitaemia OR parasitemia OR PfHRP2 OR "histidine-rich protein 2" OR "severe malaria" OR "cerebral malaria" OR "severe malarial anaemia" OR "severe malarial anemia" OR "parasite biomass").ti,ab,kw.
6. 4 OR 5
7. exp thalassemia/ OR exp glucose 6 phosphate dehydrogenase deficiency/ OR exp blood group/ OR exp erythrocyte/
8. ("red blood cell polymorphism" OR "erythrocyte polymorphism" OR "red cell polymorphism" OR "alpha thalassaemia" OR "alpha thalassemia" OR "alpha-plus-thalassaemia" OR "alpha-plus-thalassemia" OR G6PD OR "glucose-6-phosphate dehydrogenase" OR "haemoglobin C" OR "hemoglobin C" OR HbC OR ABO OR "blood group O" OR Dantu OR epistasis OR rosetting OR cytoadherence OR "parasite invasion").ti,ab,kw.
9. 7 OR 8
10. ("severe anaemia" OR "severe anemia" OR anaemia OR anemia OR transfusion OR "blood transfusion" OR "respiratory distress" OR "acute chest syndrome" OR fever OR infection OR "febrile illness" OR "malaria-like illness" OR "clinical phenotype" OR "paediatric admission" OR "pediatric admission" OR "hospital admission").ti,ab,kw.
11. ("newborn screening" OR "neonatal screening" OR "infant screening" OR "point-of-care testing" OR "point of care testing" OR "confirmatory diagnosis" OR counselling OR counseling OR "caregiver notification" OR hydroxyurea OR hydroxycarbamide OR penicillin OR prophylaxis OR vaccination OR immunisation OR immunization OR referral OR retention OR "linkage to care" OR "continuum of care" OR "comprehensive care").ti,ab,kw.
12. (underdiagnosis OR under-recognition OR "under recognition" OR "missed diagnosis" OR "clinical recognition" OR misclassification OR attribution OR "routine data" OR "health information system" OR DHIS2 OR HMIS OR coding OR reporting OR "mortality attribution" OR "cause of death" OR "service-visible burden" OR surveillance).ti,ab,kw.
13. (syndemic OR syndemics OR "structural vulnerability" OR "structural violence" OR "health system" OR "health systems" OR "service readiness" OR "spatial prioritisation" OR "spatial prioritization" OR "hub and spoke" OR "facility readiness" OR "access to care" OR "implementation science" OR "programme response" OR "program response").ti,ab,kw.
14. exp Africa south of the Sahara/
15. ("sub-Saharan Africa" OR "sub Saharan Africa" OR SSA OR Africa OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cameroon OR "Cape Verde" OR "Cabo Verde"

e" OR "Central African Republic" OR Chad OR Comoros OR Congo OR "Democratic Republic of Congo" OR DRC OR "Republic of Congo" OR "Côte d'Ivoire" OR "Cote d'Ivoire" OR Djibouti OR Eritrea OR Eswatini OR Ethiopia OR Gabon OR Gambia OR Ghana OR Guinea OR "Guinea-Bissau" OR Kenya OR Lesotho OR Liberia OR Madagascar OR Malawi OR Mali OR Mauritania OR Mauritius OR Mozambique OR Namibia OR Niger OR Nigeria OR Rwanda OR "Sao Tome and Principe" OR Senegal OR Seychelles OR "Sierra Leone" OR Somalia OR "South Africa" OR "South Sudan" OR Sudan OR Tanzania OR Togo OR Uganda OR Zambia OR Zimbabwe).ti,ab,kw.

16. 14 OR 15

17. 3 AND (6 OR 9 OR 10 OR 11 OR 12 OR 13) AND 16

18. 6 AND 9 AND 16

19. 17 OR 18

6. Scopus search strategy

Database: Scopus

Date searched: To be inserted

Records retrieved: To be inserted

TITLE-ABS-KEY (

(
"sickle cell disease" OR "sickle cell anaemia" OR "sickle cell anemia" OR SCD OR SCA OR HbSS OR HbAS OR "sickle cell trait" OR "haemoglobin S" OR "hemoglobin S" OR "Hb S" OR "sickle haemoglobin" OR "sickle hemoglobin" OR haemoglobinopathy OR hemoglobinopathy
)

AND

(
malaria OR "Plasmodium falciparum" OR falciparum OR parasitaemia OR parasitemia OR Pf HRP2 OR "histidine-rich protein 2" OR "severe malaria" OR "cerebral malaria" OR "severe malarial anaemia" OR "severe malarial anemia"

OR "red blood cell polymorphism" OR "erythrocyte polymorphism" OR "alpha thalassaemia" OR "alpha thalassemia" OR G6PD OR "glucose-6-phosphate dehydrogenase" OR "haemoglobin C" OR "hemoglobin C" OR HbC OR ABO OR Dantu OR epistasis

OR "severe anaemia" OR "severe anemia" OR transfusion OR "respiratory distress" OR "malaria-like illness"

OR "newborn screening" OR "neonatal screening" OR "point-of-care testing" OR "confirmatory diagnosis" OR hydroxyurea OR hydroxycarbamide OR referral OR retention OR "linkage to care"

OR underdiagnosis OR under-recognition OR "missed diagnosis" OR "routine data" OR "health information system" OR DHIS2 OR HMIS OR "mortality attribution"

OR syndemic OR syndemics OR "structural vulnerability" OR "service readiness" OR "facility readiness" OR "programme response" OR "program response"

)

AND

(

"sub-Saharan Africa" OR "sub Saharan Africa" OR SSA OR Africa OR Angola OR Benin OR

Botswana OR "Burkina Faso" OR Burundi OR Cameroon OR Ghana OR Kenya OR Malawi OR
Mali OR Nigeria OR Tanzania OR Uganda OR Zambia OR Zimbabwe

)
)

7. Web of Science core collection search strategy

Database: Web of Science Core Collection

Date searched: To be inserted

Records retrieved: To be inserted

TS=(

(
"sickle cell disease" OR "sickle cell anaemia" OR "sickle cell anemia" OR SCD OR SCA OR
HbSS OR HbAS OR "sickle cell trait" OR "haemoglobin S" OR "hemoglobin S" OR "Hb S" OR
"sickle haemoglobin" OR "sickle hemoglobin" OR haemoglobinopathy OR hemoglobinopathy
)

AND

(
malaria OR "Plasmodium falciparum" OR falciparum OR parasitaemia OR parasitemia OR Pf
HRP2 OR "histidine-rich protein 2" OR "severe malaria" OR "cerebral malaria" OR "severe mal
arial anaemia" OR "severe malarial anemia"

OR "red blood cell polymorphism" OR "erythrocyte polymorphism" OR "alpha thalassaemia"
OR "alpha thalassemia" OR G6PD OR "glucose-6-phosphate dehydrogenase" OR "haemoglobin
C" OR "hemoglobin C" OR HbC OR ABO OR Dantu OR epistasis

OR "severe anaemia" OR "severe anemia" OR transfusion OR "respiratory distress" OR "mala
ria-like illness"

OR "newborn screening" OR "neonatal screening" OR "point-of-care testing" OR "confirmato
ry diagnosis" OR hydroxyurea OR hydroxycarbamide OR referral OR retention OR "linkage to c
are"

OR underdiagnosis OR under-recognition OR "missed diagnosis" OR "routine data" OR "healt
h information system" OR DHIS2 OR HMIS OR "mortality attribution"

OR syndemic OR syndemics OR "structural vulnerability" OR "service readiness" OR "facilit
y readiness" OR "programme response" OR "program response"

)

AND

(
"sub-Saharan Africa" OR "sub Saharan Africa" OR SSA OR Africa OR Angola OR Benin OR
Botswana OR "Burkina Faso" OR Burundi OR Cameroon OR Ghana OR Kenya OR Malawi OR
Mali OR Nigeria OR Tanzania OR Uganda OR Zambia OR Zimbabwe
)

)
)

8. Global Health search strategy

Database: Global Health

Platform: Ovid or EBSCO, to be inserted

Date searched: To be inserted

Records retrieved: To be inserted

The search syntax should be adapted to the platform used.

1. ("sickle cell disease" OR "sickle cell anaemia" OR "sickle cell anemia" OR SCD OR SCA OR HbSS OR HbAS OR "sickle cell trait" OR "haemoglobin S" OR "hemoglobin S" OR "Hb S" OR haemoglobinopathy OR hemoglobinopathy).ti,ab,kw.

2. (malaria OR "Plasmodium falciparum" OR falciparum OR parasitaemia OR parasitemia OR P fHRP2 OR "severe malaria" OR "cerebral malaria" OR "severe malarial anaemia" OR "severe malarial anemia").ti,ab,kw.

3. ("red blood cell polymorphism" OR "erythrocyte polymorphism" OR "alpha thalassaemia" OR "alpha thalassemia" OR G6PD OR "glucose-6-phosphate dehydrogenase" OR "haemoglobin C" OR "hemoglobin C" OR HbC OR ABO OR Dantu OR epistasis).ti,ab,kw.

4. ("severe anaemia" OR "severe anemia" OR transfusion OR "blood transfusion" OR "respiratory distress" OR "acute chest syndrome" OR fever OR infection OR "febrile illness" OR "malaria-like illness").ti,ab,kw.

5. ("newborn screening" OR "neonatal screening" OR "point-of-care testing" OR "confirmatory diagnosis" OR counselling OR counseling OR hydroxyurea OR hydroxycarbamide OR penicillin OR prophylaxis OR vaccination OR immunisation OR immunization OR referral OR retention OR "linkage to care" OR "comprehensive care").ti,ab,kw.

6. (underdiagnosis OR under-recognition OR "missed diagnosis" OR "routine data" OR "health information system" OR DHIS2 OR HMIS OR coding OR reporting OR "mortality attribution" OR surveillance).ti,ab,kw.

7. (syndemic OR syndemics OR "structural vulnerability" OR "health system" OR "service readiness" OR "facility readiness" OR "access to care" OR "programme response" OR "program response").ti,ab,kw.

8. ("sub-Saharan Africa" OR "sub Saharan Africa" OR SSA OR Africa OR Angola OR Benin OR Botswana OR "Burkina Faso" OR Burundi OR Cameroon OR Ghana OR Kenya OR Malawi OR Mali OR Nigeria OR Tanzania OR Uganda OR Zambia OR Zimbabwe).ti,ab,kw.

9. 1 AND (2 OR 3 OR 4 OR 5 OR 6 OR 7) AND 8

10. 2 AND 3 AND 8

11. 9 OR 10

9. CINAHL search strategy

Database: CINAHL

Platform: EBSCOhost

Date searched: To be inserted

Records retrieved: To be inserted

(
MH "Anemia, Sickle Cell+"
OR MH "Sickle Cell Trait"
OR "sickle cell disease"
OR "sickle cell anaemia"
OR "sickle cell anemia"
OR SCD
OR SCA
OR HbSS
OR HbAS
OR "sickle cell trait"
OR "haemoglobin S"
OR "hemoglobin S"

)
AND

(
MH "Malaria+"
OR malaria
OR "Plasmodium falciparum"
OR falciparum
OR parasitaemia
OR parasitemia
OR PfHRP2
OR "severe malaria"
OR "red blood cell polymorphism"
OR "erythrocyte polymorphism"
OR "alpha thalassaemia"
OR "alpha thalassemia"
OR G6PD
OR "haemoglobin C"
OR "hemoglobin C"
OR HbC
OR ABO
OR Dantu
OR epistaxis
OR "severe anaemia"
OR "severe anemia"
OR transfusion
OR "respiratory distress"
OR "newborn screening"
OR "point-of-care testing"
OR "confirmatory diagnosis"
OR hydroxyurea
OR referral
OR retention
OR "linkage to care"

OR underdiagnosis
 OR under-recognition
 OR "missed diagnosis"
 OR "routine data"
 OR "health information system"
 OR syndemic
 OR "structural vulnerability"
 OR "service readiness"
)
 AND
 (
 MH "Africa South of the Sahara+"
 OR "sub-Saharan Africa"
 OR "sub Saharan Africa"
 OR SSA
 OR Africa
 OR Ghana
 OR Kenya
 OR Nigeria
 OR Tanzania
 OR Uganda
 OR Zambia
 OR Zimbabwe
)

10. African Index Medicus search strategy

Database: African Index Medicus

Date searched: To be inserted

Records retrieved: To be inserted

Because African Index Medicus may support simpler search syntax than bibliographic databases such as PubMed or Embase, searches should be conducted in smaller concept combinations.

Search number	Search string	Records retrieved
AIM 1	“sickle cell disease” AND malaria	To be inserted
AIM 2	“sickle cell anaemia” AND malaria	To be inserted
AIM 3	“sickle cell anemia” AND malaria	To be inserted
AIM 4	“sickle cell trait” AND malaria	To be inserted
AIM 5	“haemoglobin S” AND malaria	To be inserted

AIM 6	“hemoglobin S” AND malaria	To be inserted
AIM 7	“sickle cell disease” AND “newborn screening”	To be inserted
AIM 8	“sickle cell disease” AND hydroxyurea	To be inserted
AIM 9	“sickle cell disease” AND transfusion	To be inserted
AIM 10	“sickle cell disease” AND “severe anaemia”	To be inserted
AIM 11	“sickle cell disease” AND “routine data”	To be inserted
AIM 12	“sickle cell disease” AND syndemic	To be inserted

11. Cochrane Library search strategy

Database: Cochrane Library

Date searched: To be inserted

Records retrieved: To be inserted

("sickle cell disease" OR "sickle cell anaemia" OR "sickle cell anemia" OR SCD OR SCA OR HbSS OR HbAS OR "sickle cell trait" OR "haemoglobin S" OR "hemoglobin S")

AND

(malaria OR "Plasmodium falciparum" OR "severe malaria" OR "severe anaemia" OR "severe anemia" OR transfusion OR "newborn screening" OR hydroxyurea OR hydroxycarbamide OR penicillin OR vaccination OR "linkage to care")

AND

("sub-Saharan Africa" OR Africa OR Ghana OR Kenya OR Nigeria OR Tanzania OR Uganda OR Zambia OR Zimbabwe)

12. Grey Literature search strategy

Grey literature will be searched to capture policy documents, programme reports, implementation guidance, newborn screening initiatives, SCD service documents, and regional health-system evidence not indexed in bibliographic databases.

12.1 Grey Literature sources

Source	Website or platform	Search approach	Records identified
World Health Organization	WHO website	Site search and document search	To be inserted
Africa CDC	Africa CDC website	Site search	To be inserted

SickleInAfrica	SickleInAfrica website	Project and publication search	To be inserted
American Society of Hematology	ASH and CONSA resources	Project and report search	To be inserted
Ministries of Health in SSA countries	Country websites	Targeted search for SCD, newborn screening, malaria, child health documents	To be inserted
Global Burden of Disease	IHME or Lancet pages	SCD and malaria burden reports	To be inserted
Institutional repositories	University and research institute repositories	Keyword search	To be inserted
Conference proceedings	Relevant SCD, malaria, haematology, and public health meetings	Abstract search where accessible	To be inserted

12.2 Grey Literature search terms

The following search terms will be used singly and in combination:

1. “sickle cell disease” AND Africa
2. “sickle cell disease” AND “newborn screening” AND Africa
3. “sickle cell disease” AND malaria AND Africa
4. “sickle cell anaemia” AND “severe anaemia” AND Africa
5. “sickle cell disease” AND hydroxyurea AND Africa
6. “sickle cell disease” AND transfusion AND Africa
7. “sickle cell disease” AND “linkage to care” AND Africa
8. “sickle cell disease” AND “point-of-care testing” AND Africa
9. “sickle cell disease” AND “health system” AND Africa
10. “sickle cell disease” AND “routine data” AND Africa

11. “sickle cell disease” AND “service readiness” AND Africa

12. “sickle cell disease” AND syndemic

13. malaria AND “haemoglobin S” AND Africa

14. malaria AND “sickle cell trait” AND Africa

15. malaria AND “red blood cell polymorphism” AND Africa

12.3 Grey Literature search log template

Source	Date searched	Search terms used	Number identified	Number screened	Number included	Notes
WHO	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted
Africa CDC	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted
SickleIn Africa	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted
Ministry of Health websites	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted
Institutional repositories	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted
Conference proceedings	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted	To be inserted

13. Citation searching

Backward and forward citation searching will be conducted for key studies and reports. Citation searching will include:

1. Reference list screening of included studies
2. Reference list screening of relevant reviews
3. Forward citation searching using Google Scholar, Scopus, Web of Science, or PubMed where available
4. Related article searching in PubMed for key papers

13.1 Key seed articles for citation searching

Topic	Seed article or source	Citation search completed	Records identified
HbAS protection against malaria	Williams et al., 2005	To be inserted	To be inserted
Global HbS distribution	Piel et al., 2010	To be inserted	To be inserted
SCA birth burden mapping	Piel et al., 2013	To be inserted	To be inserted
SCD mortality in Africa	Grosse et al., 2011	To be inserted	To be inserted
Severe anaemia and SCA under-recognition	Uyoga et al., 2022	To be inserted	To be inserted
Newborn screening in Africa	Ohene-Frempong et al., 2008	To be inserted	To be inserted
Uganda sickle surveillance	Ndeezi et al., 2016	To be inserted	To be inserted
Hydroxyurea in SSA	Tshilolo et al., 2019	To be inserted	To be inserted
CONSA	Green et al., 2022	To be inserted	To be inserted
Syndemic theory	Singer and Clair, 2003	To be inserted	To be inserted
Syndemic methods critique	Tsai, 2018	To be inserted	To be inserted

14. Search documentation and reproducibility

For each database and grey literature source, the review team will document:

1. Source name
2. Platform or interface
3. Date searched
4. Full search string
5. Filters or limits applied
6. Number of records retrieved

7. Export format
8. Deduplication process
9. Number of records after deduplication

The search strategy will be updated only if necessary. Any changes after protocol registration will be recorded in the protocol amendment log.

15. Notes on search sensitivity and specificity

The search strategy was designed to be sensitive because the review spans multiple disciplines and evidence types. The strategy uses broad SCD and malaria terms combined with red blood cell polymorphism, clinical, service visibility, screening, and syndemic terms. Supplementary searches were included to capture SCD screening and care literature in SSA that may not explicitly mention malaria but remains relevant to the malaria-SCD syndemic framework because it addresses service-mediated visibility, structural vulnerability, or programme response in malaria-endemic settings.

Final screening decisions will determine relevance to the review question.